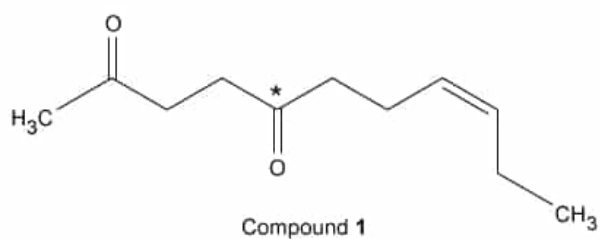
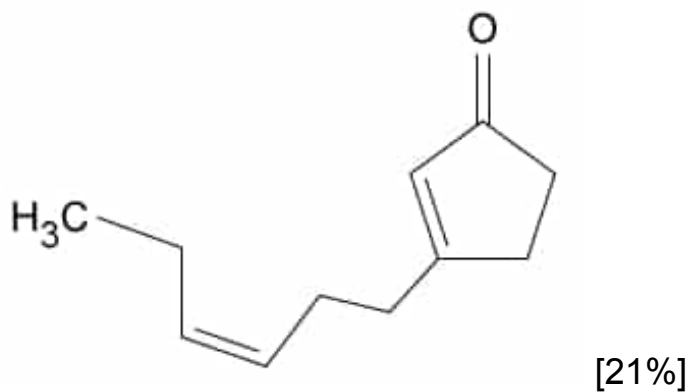


Compound **1** shown below is converted to Compound **2** by a base-catalyzed aldol condensation.

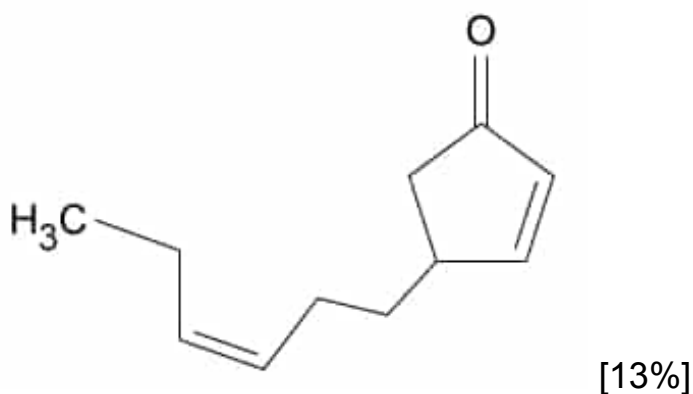


If the carbonyl labeled with an asterisk is the carbonyl in the final product, what is the structure of Compound **2**?

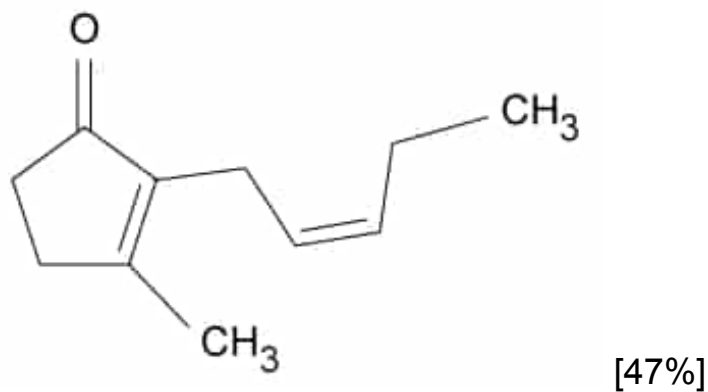
☐ A.



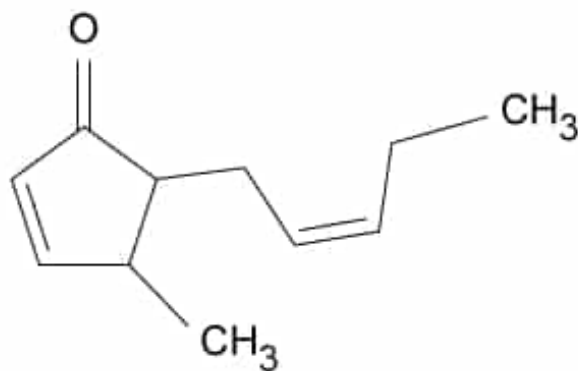
☐ B.



✓ ☒ C.



☐ D.

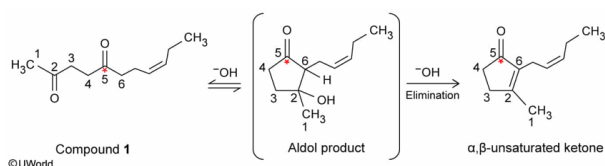


[17%]

Correct

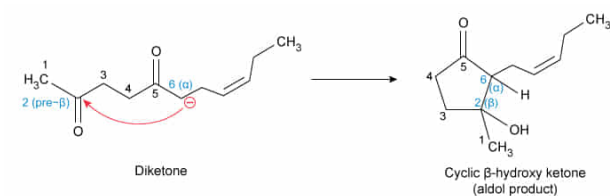
48% answered correctly

Explanation:

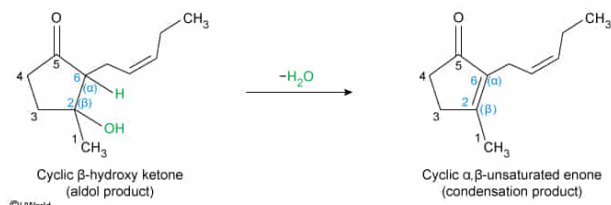


An **aldol condensation** requires two carbonyl groups (from ketones and/or aldehydes), and these may be from separate molecules or within the same molecule. The reaction begins with deprotonation of an **α -carbon** (carbon adjacent to a carbonyl) to form an enolate. Next, the nucleophilic enolate attacks another carbonyl, forming a new **carbon-carbon bond** (the aldol product). The carbonyl *adjacent* to the nucleophile remains intact, and the carbonyl that *received* the nucleophile is reduced to a hydroxyl. Finally, -OH elimination takes place to give an **α,β -unsaturated enone**. A reaction between two separate compounds results in the addition of one compound to the other, whereas an intramolecular reaction within the same molecule (as in a **diketone**) results in a **cyclization**.

Compound **1** is a 1,4-diketone and forms a five-membered ring in an aldol condensation. Because the intact carbonyl in the α,β -unsaturated product is on carbon 5, the nucleophilic α -carbon must have been either carbon 4 or 6. It is energetically unfavorable for carbon 4 to be the nucleophile because a strained three-membered ring would form, whereas a favorable five-membered ring forms if carbon 6 is the nucleophile.



After nucleophilic attack, carbon 2 is reduced from a **carbonyl carbon** to the **β -hydroxylated carbon** of the aldol reaction product. Subsequent deprotonation of carbon 6 results in -OH elimination from carbon 2 and the formation of a double bond between carbons 2 and 6 to yield the condensation product, an α,β -unsaturated ketone.



(Choices A and B) These structures form if carbon 1 acts as the nucleophile rather than carbon 6, leaving carbon 2, rather than carbon 5, with the carbonyl.

(Choice D) The alkyl substituents are on the correct carbon atoms in this structure; however, the double bond in the ring is on the incorrect side of the carbonyl. Because the double bond is a result of -OH elimination, the double bond should be between the carbon with acidic protons (carbon 6) and the carbon with the -OH leaving group (carbon 2).

Educational objective:

An aldol condensation results in a new carbon–carbon bond and an α,β -unsaturated enone. The reaction requires two carbonyl groups (from ketones and/or aldehydes) and occurs by the nucleophilic attack of an enolate on a carbonyl followed by elimination. A diketone substrate will result in cyclization and formation of a five- or six-membered ring.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401762

System:

Functional Groups and Their Reactions